

Why did you choose your proposed course and institution?

Please include information on efforts you have undertaken so far to obtain information in your study option.

I have chosen to pursue a Master of Artificial Intelligence at Monash University because it's a pivotal role in shifting my career into an AI engineer while enhancing my expertise in initiating innovation to serve a safer digital environment for society through AI. This passion stems from a deep concern in Indonesia, where over 12,500 hoax articles have misled the public in the past five years. Being the fourth in number of social media users in the world and projected to have about 275 million users (almost 95% of the total population) in 2029 could increase the possibility of misinformation spreading and growing rapidly more than ever. With almost 60% of Indonesian people being exposed to hoaxes, mostly on political, government, and health-related issues, this makes it a serious problem to be mitigated.

Working at a public technology company has allowed me to see firsthand how emerging AI can provide transformative solutions. Consequently, I became driven to assist my country in addressing this issue by inventing AI solutions that could detect and filter false narratives. Choosing Monash University was a heartfelt decision, they offer a curriculum that combines Large Language Models (LLMs) as the basis to analyse huge textual data and the study of research and innovation in AI with an emphasis on social good, which ultimately provides the ideal environment to transform my passion into tangible solutions.

I also reached out to lecturers from Monash and discovered that Professor Thanh Thi Nguyen was working on leveraging LLMs to combat misinformation, which further solidified my decision to study at Monash. Interestingly, I have the opportunity to intern at the AI for Law Enforcement and Community Safety (AiLECS) Lab, that has a partnership with the Australian Federal Police. This internship enabled me to participate in sustainable AI projects focused on identifying misinformation in public and establishing a worldwide network with like-minded researchers.

What impact do you believe this study will have on your career, life and community?

Studying Artificial Intelligence (AI) at Monash University will surely help me broaden my influence on society while strengthening my career path and personal growth. Professionally, this programme will equip me with advanced knowledge to leverage LLMs by learning multi-agent systems and automated reasoning. This knowledge is a crucial material for me to lead the development of a cognitively intelligent language model that runs effortlessly on mobile devices, automatically filtering false information and alerting users to harmful content. With the support from Telkom Indonesia, where I am currently working, my goal is to launch this tool within two years after completing my studies in Australia.

On a personal level, engaging with Australia's innovative AI research environment will develop my strategic thinking and greatly improve my technical skills. I believe that collaborating with global experts, including Professor Thanh Thi Nguyen, will further expand my professional network and initiate meaningful partnerships, crucial for promoting and integrating my technological solutions effectively. Moreover, about 93% of the alumni successfully established new networks, and 71% of them managed to leverage the networks to contribute to Indonesia's development (AAI Survey, 2024). This convinced me that my studies will not only enhance personal traits, but they will also serve as a catalyst for meaningful change in the greater community. My goal is to reduce the efficacy of growing misinformation in Indonesia by 50%

within three years after I fully develop the tools. To achieve this, I am also going to conduct online advocacy campaigns, especially for people in remote areas, targeting 3000 individuals within three years and triggering a ripple effect.

How have you contributed to solving a challenge or implementing change or reform in the context of one of the AAI priority development areas?

(Be specific and include: what aspect/s of your leadership knowledge, skills and practice you consider to be well established and effective; which people or organisations you worked with to solve the problem, and what creative methods were used.)

In 2024, Telkom Indonesia suffers from a potential loss of about 10% of the total revenue caused by fraudulent transactions within the company. As a data scientist at the Internal Data Analytics unit, I have the opportunity to lead and generate solutions to tackle this challenge. I initiated a team that consists of five internal auditors with none of them having a STEM background, to develop a fraud alert system by leveraging machine learning. At first, it was difficult to align the objectives with the knowledge gap in the group; hence, I nurtured them starting with the fundamentals of data science. This enabled me to assist them in building their foundational skills and developing their digital leadership.

To ensure this project finishes according to plan, I organise a weekly sharing session with the team to consolidate their knowledge and fill their gaps. Consequently, they set me as their role model and began to deepen their expertise independently by attending online classes and seminars that I suggested, enabling me to work on the project seamlessly. After four months, I managed to launch the tools capable of automatically reporting any fraudulent transactions within the company as well as digitalise the current process that has run manually for years, namely "CLUDIA". By implementing this tool, I was able to assist Telkom Indonesia in saving around AUD 75 million in potential revenue loss by fraudulent acts.

Finally, in January 2025, the Corruption Eradication Commission (KPK) successfully detained three corruption suspects by working closely with the internal auditors. The higher-ups also recognise my efforts by giving me direct appreciation, especially my Executive Vice President, Mr Komang Budi. This experience taught me to immerse myself in giving impact to society, and I am very pleased in the process. It also motivates me to scale my contribution to an even greater community after my studies in Australia.

Please provide three practical and realistic examples of how you intend to use the knowledge, skills and connections you will gain from your scholarship on your return to Indonesia. Possible tasks can be personal and/or professional.

Upon my return to Indonesia, I plan to channel the knowledge, skills, and networks gained from studying in Australia into practical initiatives focused on mitigating misinformation by leveraging AI.

First, with the support from Telkom Indonesia, I am going to shift my role to become an AI Engineer, applying my Australia-acquired expertise and gaining access to industry-level AI use cases. Also, by leveraging Telkom Digital Amoeba, a corporate innovation lab that offers infrastructure, I aim to actualise my research on Large Language Models for misinformation detectors that run effortlessly on mobile devices. In doing so, I am planning to initiate a task force consisting of experts at Telkom and supervised by external academics from my studies to refine this invention.

Second, I am aware that public awareness is one of the keys to fighting misinformation. Therefore, while developing the tool, I am going to initiate a programme by utilising the AAI Alumni Network to host quarterly online seminars to improve AI literacy. Especially focusing on how the digital era can both help and harm information delivery and how the emerging AI and its ethics can also intellectualise the people to filter out the false contents. My goal is to reach 3000 individuals within three years and trigger ripple effects by prioritising communities in remote areas that have a low digital safety index.

Lastly, with the tool and ecosystem ready, I plan to introduce this invention to the government level. I am going to initiate collaboration with INA Digital as Indonesia's government initiative in advancing digital technologies. By targeting two years after my studies, I hope to assist the country in creating a safer environment for society where Indonesia's digital era reaches its peak in 2030. Through these initiatives, I directly bridge Indonesia and Australia by participating in collaborative strategies to strengthen institutions and hopefully addressing challenges in both countries.

List any possible constraints you think may prevent you from achieving above tasks.

First, the complexity and resource intensity of developing AI solutions could hinder large-scale implementation. To address this, I am going to leverage my position as a member of the Telkom Data Science and AI task force to gain significant support and currently conduct the initial development. Furthermore, I am aiming to develop this into a Minimum Viable Product (MVP) during my studies in Australia. Assisted by professionals and academia, I will gain important feedback and trigger continuous improvement. Additionally, I design a clear roadmap and milestones ensuring the project's execution.

The next challenge is optimising the AI advocacy programme through online seminars, particularly to reach communities in remote areas, such as eastern Indonesia, where internet penetration remains below the national average (70.6% as of 2023). Therefore, I am going to collaborate closely with the AAI alumni network to leverage their extensive community connections and resources. It will allow me to provide alternative methods, such as offline-accessible recordings or culturally relevant materials in local languages. Moreover, I am planning to gain sponsorship from Telkom Indonesia to provide free internet packages for eligible participants, fostering engagement.

Lastly, bureaucratic and regulatory barriers might delay government-level adoption of innovative technologies. To anticipate this challenge, I am going to organise a sharing session with Telkom's team which develops government digital projects, such as GovTech Procurement, Satu Sehat, and PaDi UMKM. This is a crucial step in gaining knowledge for further seamless collaboration with the INA Digital and facilitating smoother AI tools implementation at the

nationwide level. I am certain that by taking proactive measures to address these challenges, I can drive long-term change in mitigating the widespread misinformation in public, contributing to strengthening Indonesia's institutions through AI.